Stability of the regular n-gon of point vortices at its bifurcation point: completing the proof of the Montaldi–Tokieda conjecture

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Abstract

Montaldi and Tokieda studied a one-parameter family of point-vortex systems on the sphere, the plane and the hyperbolic plane, and showed that a regular ring of $n \geq 4$ identical vortices loses stability at an explicit bifurcation value b_n of the curvature parameter. They conjectured that the ring is still stable *at* b_n (their Conjecture 13), in the sense of Lyapunov stability modulo rotations. They proved it for all even n, where the sign of the decisive quartic coefficient is settled by a Maple positivity check, and for $n = 3, 5, 7, 9, 11$, where the odd-n coefficients are evaluated numerically. We give the first proof for odd $n \geq 13$. Our argument is uniform in n, so it also re-proves the cases $4 \leq n \leq 11$ by a different route. Stability is Lyapunov stability modulo rotations about the ring's centre, obtained by an energy–momentum argument on the momentum level set, and it holds for all small perturbations, including those that change the momentum. The proof has three parts. (i) A spectral computation. In log-polar coordinates the reduced Hessian at b_n is block-diagonal and independent of the geometry, and its kernel is exactly the critical Fourier mode. (ii) Exact closed forms, for all n, for the quartic coefficient c_4 of the reduced Hamiltonian along that kernel, after the transverse directions are eliminated on the momentum level set: $c_4 = (n-2)(m^4+40m^2+48m+16)/(192n(n-1))$ with $m = n-2$ for even n, and $c_4 = (n-1)(n-3)(n^2-9n+38)/(128n)$ for odd $n \geq 5$. These replace Montaldi and Tokieda's Maple check (even n) and numerical checks (odd $n \leq 11$). Positivity is immediate once the formulas are known; the work is in deriving them. (iii) A self-contained energy–momentum argument for degenerate (quartic) minima. It works for the noncompact groups $SE(2)$ and $PSL(2,\mathbb{R})$, and for the plane's affine momentum map. The algebra that turns the computed series coefficients into the closed forms is checked in Lean 4. Our results confirm Montaldi and Tokieda's conclusions. All verification so far has been by AI readers and computer checks; no human specialist has yet reviewed the argument.

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1 Introduction

The family. Montaldi and Tokieda [7] put point vortices on the sphere, the plane and the hyperbolic plane into one smooth family. In a uniform chart $z \in \mathbb{C}$ they use the surface M_λ of curvature 4λ, with

$$\sigma(z) = 1 + \lambda|z|^2, \qquad \Omega_\lambda = \frac{2}{\sigma^2}\, dx \wedge dy \quad ([7,\ (1.8)]), \qquad J_\lambda(z) = \frac{1}{\sigma}\big(z, |z|^2\big) \quad ([7,\ (1.9)]),$$

and the pair interaction [7, (1.12)]

$$G(z; w) = \log \frac{|z - w|^2}{(1 + \lambda|z|^2)(1 + \lambda|w|^2)}.$$

For $\lambda > 0$ this is the standard sphere (stereographic chart). For $\lambda = 0$ it is the plane. For $\lambda < 0$ it is the hyperbolic plane with the "sinh" interaction, i.e. $G = 2\log\sinh(d/2)$ up to constants, where d is hyperbolic distance. The symmetry group G_λ is $SO(3)$, $SE(2)$ or $PSL(2, \mathbb{R})$ respectively. For n identical unit vortices at distinct points $z_1, \dots, z_n$ we use

$$H = -\sum_{i<j} \log \frac{|z_i - z_j|^2}{\sigma_i \sigma_j}, \qquad \Omega = \sum_k \frac{2}{\sigma_k^2} dx_k \wedge dy_k, \qquad J = (J_1, J_2, J_3) = \sum_k \Big(\frac{x_k}{\sigma_k}, \frac{y_k}{\sigma_k}, \frac{|z_k|^2}{\sigma_k}\Big), \quad (1)$$

with $\sigma_k = \sigma(z_k)$. This is the Hamiltonian of [7] without the positive factor $\kappa^2/4\pi$. That factor rescales time and does not affect any statement below.

Rings and the bifurcation point. The regular ring $z_k = r_0 e^{2\pi i k/n}$ is a relative equilibrium for every r_0 and λ. By scaling we may take $r_0 = 1$, so the only parameter is $b = \lambda r_0^2 \in (-1, 1]$. Montaldi–Tokieda [7, Thm. 6] prove that for $n \geq 4$ the ring is stable for $\lambda r_0^2 < b_n$, where b_n is the unique root in $(-1, 1]$ of

$$\frac{1 + b_n^2}{(1 + b_n)^2} = \frac{1}{2(n-1)} \Big\lfloor \frac{n^2}{4} \Big\rfloor \qquad ([7,\ (3.6)]). \quad (2)$$

Above b_n the ring is spectrally unstable, and a pitchfork bifurcation occurs. Explicitly,

$$b_n = \frac{4(n-2)\sqrt{n-1} - n^2}{n^2 - 8n + 8} \ \ (n \text{ even}), \qquad b_n = \frac{4\sqrt{n-3} - (n+1)}{n - 7} \ \ (n \text{ odd},\ n \neq 7), \qquad b_7 = 0.$$

So $b_4 = 2 - \sqrt{3}$, b_5 and $b_6 = 9 - 4\sqrt{5}$ are positive (sphere). The value $b_7 = 0$ is the plane, where this is the Thomson heptagon. For $n \geq 8$, $b_n < 0$ (hyperbolic plane); for example $b_8 = -(8 - 3\sqrt{7})$. Throughout, $|b_n| < 1$.

The conjecture. Page numbers for [7] refer to arXiv:1210.5662v1. Montaldi and Tokieda define stability as follows [7, p. 10]: "The precise sense of stability we are adopting is like the ordinary one of Lyapunov stability, but in terms of G_μ-invariant open sets, where μ is the momentum value at the trajectory and G_μ is the stabilizer of μ: in detail, for every G_μ-invariant neighbourhood V of the trajectory, there exists a G_μ-invariant neighbourhood $U \subseteq V$ such that every trajectory starting in U remains in V for all time." Before Theorem 6 they add [7, p. 15]: "Stability means Lyapunov stability modulo rotations (same as orbital stability in our situation)." Then, on p. 23:

> **Conjecture 13.** *For all values of n, the n-gon is stable at the bifurcation point.*
> Above, we have proved this for all even n and for $n = 3, 5, 7, 9, 11$.

The notion of stability. Throughout, "stable" means Lyapunov stability modulo rotations about the ring's centre, in the sense of [7, p. 10, p. 15] just quoted (Theorem 1 states it precisely). We obtain it by the energy–momentum method: the reduced Hamiltonian on the momentum level set $J^{-1}(\mu)/K$ has a strict, but degenerate (quartic), local minimum, and Section 5 turns this into Lyapunov stability modulo K in the full phase space.

What was known. Montaldi and Tokieda [7] proved Conjecture 13 for all even n and for $n = 3, 5, 7, 9, 11$, as quoted above. For even n [7, §3.5.1, pp. 20–21], the cubic term vanishes on the slice (their Lemma 10), and the t^4 coefficient along the critical mode is a positive multiple of a bracket T. Substituting the two roots of the threshold relation gives two values T_1, T_2, and closed forms for $T_1 + T_2$ and $T_1 T_2$, both manifestly positive for $m = n - 2 \geq 0$, are obtained "using Maple!" (p. 21). For odd n [7, §3.5.2, pp. 22–23], the criterion involves three coefficients β, γ, δ. The coefficient β has a closed form, while "the awkward trigonometric expressions prevented us (and Maple) from reaching closed forms for γ and δ" (p. 23). The coefficients are evaluated numerically for $n = 5, 9, 11$. For $n = 7$ they are found with Maple as rational multiples of $1/\pi r_0^k$, which are "correct to a high degree of precision—but we have no proof that they are rational multiples of $1/\pi r_0^k$"; the sign of $\beta\delta - \gamma^2$ is then certain. Their case $n = 3$ rests on their statement (p. 15) that "the ring of $n = 3$ vortices is always nonlinearly stable". So odd $n \geq 13$ (all in the hyperbolic plane) remained open. The case $n = 7$ is the Thomson heptagon. Its nonlinear stability was proved by Kurakin and Yudovich [4] (we have read only the abstract) and by Schmidt [9]; Montaldi–Tokieda recover it [7, Cor. 7]. For the sphere at the critical latitude ($n = 4, 5, 6$), Borisov and Kilin [1] wrote that the problem "requires separate analysis with respect to nonlinear terms and remains unsolved to the present day". Kurakin [2, 3] later gave necessary and sufficient conditions for nonlinear stability of regular vortex systems on the sphere. We have not read those papers and make no claim about their content.

Contributions.

1. The first proof of Conjecture 13 for odd $n \geq 13$. Together with [7] this settles the conjecture for all n. Our proof (Theorem 1) is uniform for all $n \geq 4$, so it gives an independent proof of the cases $4 \leq n \leq 11$ already proved in [7]. Stability is in Montaldi–Tokieda's sense, and it holds for perturbations of arbitrary (nearby) momentum.

2. Exact closed forms for the reduced quartic coefficient c_4, for every even and every odd n (Theorems 6 and 7). They replace the Maple positivity check of [7] for even n and their numerical evaluations for odd $n = 5, 7, 9, 11$. Once the formulas are known, positivity is immediate (Remark 8); the content is the derivation.

3. A self-contained nonlinear-stability argument for degenerate minima that covers the noncompact groups involved, including the plane with its affine momentum map (Section 5).

4. A Lean 4 check of the algebra that leads from the computed series coefficients to the closed forms and to Table 1 (Section 8, which states exactly what is and is not formalised).

2 Main result

Theorem 1. *Let $n \geq 4$. Consider n identical point vortices on M_λ with Hamiltonian (1), in a regular ring at the bifurcation point $\lambda r_0^2 = b_n$. This means the sphere for $n = 4, 5, 6$, the plane for $n = 7$ and the hyperbolic plane for $n \geq 8$. Let $K \cong SO(2)$ be the group of rotations about the ring centre, so that $K = G_\mu$ for $\mu = J(\text{ring})$. Then the ring is Lyapunov stable modulo K. That is, for every $\varepsilon > 0$ there is $\delta > 0$ such that every solution with $\operatorname{dist}(x(0), K \cdot x_0) < \delta$ exists for all $t \in \mathbb{R}$ and satisfies $\operatorname{dist}(x(t), K \cdot x_0) < \varepsilon$. No restriction is placed on the momentum of $x(0)$.*

Since K is compact, K-invariant neighbourhoods of the (compact) relative-equilibrium orbit $K \cdot x_0$ are cofinal among all of its neighbourhoods. So Theorem 1 is exactly stability in the sense of [7, p. 10] quoted above. It also implies stability modulo the full group G_λ.

Corollary 2. *Conjecture 13 of [7] holds for every $n \geq 4$. Together with the case $n = 3$, which [7] state is always nonlinearly stable [7, p. 15], it holds for all n.*

For $n = 3$, equation (2) has its only root at $b = 1$ (the equator). There the momentum value is fixed by all of $SO(3)$, the hypothesis $|b| < 1$ used throughout fails, and the critical mode is $m = 1$, which meets the momentum directions. Consistently with this, the odd-n formula of Theorem 7 gives $c_4 = 0$ at $n = 3$, so it cannot decide that case, and its derivation assumes $n \geq 5$ in any case. We rely on Montaldi–Tokieda's statement for $n = 3$ and do not re-prove it. For $n \geq 4$ the new content is odd $n \geq 13$; for $4 \leq n \leq 11$, Theorem 1 is an independent proof of results already in [7].

Structure of the proof. Section 3 shows that the reduced Hessian at b_n is positive semidefinite, with kernel equal to the radial part of the critical Fourier mode (Proposition 4). Section 4 computes the quartic coefficient of the reduced Hamiltonian along this kernel, after the transverse directions are eliminated, and shows it is positive (Theorems 6 and 7). Section 5 turns "degenerate strict minimum of the reduced Hamiltonian" into Lyapunov stability modulo K in the full phase space (Theorem 17).

3 Setup: Fourier blocks and the reduced space

3.1 Log-polar coordinates and the Fourier blocks

Write $z_k = e^{s_k + i(\phi_k + \theta_k)}$ with $\phi_k = 2\pi k/n$, so the ring is $s = \theta = 0$. Then

$$\log |z_j - z_k|^2 = s_j + s_k + \log\big(2\cosh(s_j - s_k) - 2\cos(\phi_j - \phi_k + \theta_j - \theta_k)\big),$$

which does not involve λ. All dependence on λ is in one-body terms that depend on the s_k alone. These are $(n-1)\sum_k \log(1 + b\,e^{2s_k})$ in H, and $J_3 = \sum_k g(e^{2s_k})$ with $g(u) = u/(1+bu)$.

Let $F = H - \omega_0 J_3$, with ω_0 chosen so that $dF(\text{ring}) = 0$. With $D_\phi = 2 - 2\cos\phi$, the pair logarithm has no linear Δs term and no $\Delta s\,\Delta\theta$ term at second order; its quadratic part is $(\Delta s^2 -$

$\Delta\theta^2)/D_\phi$. The Fejér-type identity

$$\sum_{l=1}^{n-1} \frac{1-\cos(2\pi ml/n)}{1-\cos(2\pi l/n)} = m(n-m) \qquad (0 \le m \le n) \tag{3}$$

(equivalent to [7, (2.4)]) diagonalises the pair part on each Fourier mode. Its proof is one line: with $w = e^{2\pi il/n}$ the summand is $|1 + w + \cdots + w^{m-1}|^2$, then sum over l.

The one-body part of F is $\sum_k \varphi(s_k)$ with $\varphi(s) = -(n-1)s + (n-1)\log(1+be^{2s}) - \omega_0\, e^{2s}/(1+be^{2s})$. The condition $\varphi'(0) = 0$ gives $\omega_0 = -(n-1)(1-b^2)/2$, and then

$$C_1 := \varphi''(0) = \frac{4(n-1)b}{(1+b)^2} + \frac{(n-1)(1-b^2)}{2} \cdot \frac{4(1-b)}{(1+b)^3} = \frac{2(n-1)(1+b^2)}{(1+b)^2}.$$

Let V_m ($0 \le m \le \lfloor n/2 \rfloor$) be the real Fourier spaces, spanned by $(\cos m\phi_k, \sin m\phi_k)$ in s and in θ. The space V_m has dimension 4 for $0 < m < n/2$, and dimension 2 for $m = 0$ and for $m = n/2$.

Proposition 3 (Fourier blocks)**.** *For every $b \in (-1, 1)$, $d^2F(ring)$ is block-diagonal over the V_m, with no radial–tangential coupling. Its eigenvalues are*

$$a_m = C_1 - m(n-m) \quad \textit{(radial)}, \qquad b_m = m(n-m) \quad \textit{(tangential)},$$

with respect to the Euclidean structure $\sum_k (ds_k^2 + d\theta_k^2)$. At $b = b_n$ we have $C_1 = \lfloor n^2/4 \rfloor = \max_m m(n-m)$, which is (2).

This is [7, (2.6)] up to normalisation. We have $a_m \ge 0$ at b_n, with equality iff $m = \lfloor n/2 \rfloor$, because $m(n-m)$ is strictly increasing for $1 \le m \le n/2$. For odd n, the modes $(n \pm 1)/2$ form the single real space $V_{(n-1)/2}$.

3.2 The reduced space

Let x_0 be the ring and $\mu = J(x_0) = (0, 0, \mu_3)$ with $\mu_3 = n/(1+b)$. The stabiliser G_μ is the group K of rotations about the origin, because $|b| < 1$ excludes the equator. Let t denote the rotation direction, $\theta_k = 1$ for all k.

(a) $dJ_3 = 2(1+b)^{-2} \sum_k ds_k$ is nonzero exactly on the radial part of V_0.

(b) Let $J_+ = J_1 + iJ_2$, and let R be the rotation by $2\pi/n$ composed with the relabelling $k \mapsto k+1$. Then R fixes the ring and $J_+ \circ R = e^{2\pi i/n} J_+$. So $dJ_+(x_0)$ vanishes on V_m for $m \ne 1$. Explicitly $dJ_+ = \sum_k e^{i\phi_k}(A\, ds_k + iB\, d\theta_k)$, with $A = (1-b)/(1+b)^2$ and $B = 1/(1+b)$. It has real rank 2 on V_1; let $V_1' = V_1 \cap \ker dJ_+$, which has dimension 2.

(c) The other two generators ξ_1, ξ_2 of G_λ are rotations about horizontal axes, translations, or boosts. They span a 2-plane in V_1 that is *not* in $\ker dJ$, because $\det\big(dJ_i(\xi_j)\big)_{i,j=1,2} = \{J_1, J_2\}(x_0)^2$ and

$$\{J_1, J_2\} = \tfrac{n}{2} - \lambda J_3, \qquad \{J_1, J_2\}(x_0) = \frac{n(1-b)}{2(1+b)} \ne 0. \tag{4}$$

For $\lambda = 0$ the bracket is the constant $n/2$. This is the central term of the affine (non-equivariant) momentum map of $SE(2)$. Our argument never uses equivariance, only that J_1, J_2, J_3 are Hamiltonians for the generators and the value (4). For $\lambda \ne 0$, $J - (0, 0, n/(2\lambda))$ is equivariant.

Hence $\ker dJ(x_0) = \mathbb{R}t \oplus V_1' \oplus V_2 \oplus \cdots \oplus V_{\lfloor n/2 \rfloor}$. The G_λ-orbit meets $\ker dJ$ only in $\mathbb{R}t$. The reduced space $M_\mu = J^{-1}(\mu)/K$ is a smooth manifold near $[x_0]$ (K acts freely on labelled rings), and

$$\dim M_\mu = 2n - 4.$$

Three dimensions are removed by constraints (the radial part of V_0, and two directions in V_1), and one by the gauge direction t.

3.3 Gap (i): spectral positivity off the kernel

Let h_μ be the reduced Hamiltonian. On $J^{-1}(\mu)$ we have $H = F + \omega_0\mu_3$. For any curve c in $J^{-1}(\mu)$ through x_0, $(F \circ c)''(0) = d^2F(c', c') + dF(c'') = d^2F(c', c')$, because $dF(x_0) = 0$. So $d^2h_\mu([x_0])$ is $d^2F(x_0)$ restricted to $\ker dJ(x_0)$, with t in its radical.

Proposition 4. *Let $n \geq 4$ and $\lambda r_0^2 = b_n$, in any of the three geometries. The Hessian of h_μ at $[x_0]$ is positive semidefinite. Its kernel is exactly the radial part of $V_{\lfloor n/2 \rfloor}$, of dimension 1 for even n and 2 for odd n. It is positive definite on the complement $E = V_1' \oplus \bigoplus_{2 \leq m \leq \lfloor n/2 \rfloor} (\text{tangential } V_m) \oplus \bigoplus_{2 \leq m < \lfloor n/2 \rfloor} (\text{radial } V_m)$.*

Proof. By Proposition 3 the restriction is block-diagonal. *Mode 1:* on all of V_1, $d^2F = \mathrm{diag}(a_1 I, b_1 I)$ with $b_1 = n - 1 > 0$ and $a_1 = \lfloor n^2/4 \rfloor - (n-1) > 0$ for $n \geq 4$. So d^2F is positive definite on every subspace of V_1, in particular on V_1'. Nothing about A or B is needed. *Modes $m \geq 2$:* $b_m > 0$, and $a_m > 0$ unless $m = \lfloor n/2 \rfloor$, where $a_m = 0$. *Mode 0:* its radial part is removed by $J_3 = \mu_3$ and its tangential part is the gauge direction. □

The smallest nonzero eigenvalue (in the (s, θ)-Euclidean normalisation) is 1 for even $n \geq 6$ ($a_{n/2-1} = 1$) and 2 for odd $n \geq 7$ ($a_{(n-3)/2} = 2$). For $n = 4, 5$ the next mode is $m = 1$, where V_1' mixes radial and tangential parts; the values are $3/2$ and $8/3$. An independent 40-digit computation of the full $2n \times 2n$ Hessian, compressed onto $\ker dJ \cap t^\perp$, agrees with these predictions for $n = 4, \ldots, 15$ to $5 \cdot 10^{-39}$. A bordered-inertia check at $n = 5, 7, \ldots, 25, 31$ finds exactly the predicted kernel and no negative eigenvalue (Section 8).

4 The effective quartic on the level set

4.1 Why we work on the level set

Let K_0 be the kernel in Proposition 4, with coordinates $(a, b) \in K_0 \times E$ on M_μ. Let $\beta(a)$ be the slaving map, the critical point of $b \mapsto h_\mu(a, b)$, and let $g(a) = h_\mu(a, \beta(a)) - h_\mu(0)$. Because the Hessian vanishes on K_0, the leading term of g is of order 4 (the cubic term vanishes by symmetry; see below). It is built from three ingredients: the quartic derivative of H along a, the cubic coupling of a^2 to the slaved $O(|a|^2)$ displacement, and the quadratic energy of that displacement.

The slaved displacement must keep the curve on $J^{-1}(\mu)$. Moving a distance ε along a kernel vector changes J_3, and through J_+ also the mode-1 momenta, at order ε^2. So a curve in the level set has an $O(\varepsilon^2)$ component along directions *not* in $\ker dJ(x_0)$, chiefly the radial part of V_0. This component has Hessian weight $a_0 = C_1 \neq 0$ and couples to a^2 at cubic order, so it contributes at order ε^4.

When the Hessian is nondegenerate, $O(\varepsilon^4)$ is negligible next to the quadratic term, and the curvature of the level set is invisible. When the Hessian is degenerate, ε^4 is the leading order, and the curvature of $J^{-1}(\mu)$ enters the decisive coefficient. For this reason we define the quartic coefficient directly on the momentum level set.

Definition 5. Let $e \in \mathbb{R}^{2n} \cong \mathbb{C}^n$ be the Euclidean-unit radial kernel vector in the chart: $e_k = n^{-1/2}(-1)^k \hat{z}_k$ for even n, and $e_k = (2/n)^{1/2}\cos(l\phi_k)\hat{z}_k$ with $l = (n-1)/2$ for odd n. Here $\hat{z}_k$ is the outward unit vector at the k-th vortex. For odd n, let e' be the same with sin in place of cos. Define

$$f_{\text{eff}}(\varepsilon) = \min\big\{H(q) - H(x_0) :\ J(q) = \mu,\ \langle q - x_0, e\rangle = \varepsilon,\\ \langle q - x_0, e'\rangle = 0\ (n \text{ odd}),\ \langle q - x_0, t\rangle = 0,\ q \text{ near } x_0\big\}.$$

Then $f_{\text{eff}}(\varepsilon) = c_4\varepsilon^4 + O(\varepsilon^5)$, and c_4 is the *level-set effective quartic*.

f_{eff} is g restricted to a ray in K_0, written in a different chart. Positivity of the quartic term does not depend on the chart: a change of chart acts on K_0 by an invertible linear map to leading order. By symmetry, g has no cubic term and its quartic term is $c_4|a|^4$:

- For even n, the generator of $\mathbb{Z}_n$ acts by -1 on $K_0 =$ radial $V_{n/2}$, so g is even.
- For odd n, $\mathbb{Z}_n$ acts on $K_0 \cong \mathbb{C}$ by $a \mapsto e^{2\pi i l/n}a$ with $\gcd(l, n) = 1$. The monomial $a^p\bar{a}^q$ is invariant iff $n \mid p - q$. For $p + q \le 4 < n$ ($n \ge 5$) this forces $p = q$. So there is no cubic, and the only quartic is $|a|^4$, as in [7, Prop. 12].

Since h_μ, J and the Fourier splitting are $\mathbb{Z}_n$-equivariant, g is $\mathbb{Z}_n$-invariant.

4.2 Even n: symmetric criticality and product formulas

Theorem 6. *For even $n = 2M \ge 4$ at $b = b_n$,*

$$c_4 = \frac{(n-2)\,(m^4 + 40m^2 + 48m + 16)}{192\,n\,(n-1)}, \qquad m = n - 2.$$

In particular $c_4 > 0$, and $c_4 = \frac{1}{4}, \frac{23}{30}, \frac{95}{56}, \frac{49}{15}, \frac{755}{132}$ for $n = 4, 6, 8, 10, 12$.

Proof. Symmetric criticality. The kernel vector e is fixed by the subgroup $D_M \subset D_n$ generated by rotation by $2\pi/M$ and the reflection fixing vortex 0. For fixed ε the slaved problem is a strictly convex minimisation near x_0 (Proposition 4 and the implicit function theorem). It is D_M-equivariant, so its unique minimiser is D_M-invariant. The level set and the constraints are also D_M-invariant. Hence the minimiser lies in $\mathrm{Fix}(D_M) \cap J^{-1}(\mu)$. Up to rotation, this set is the curve of configurations made of two staggered regular M-gons,

$$\{\rho_+\eta^j\}_{j<M} \cup \{\rho_-\zeta\eta^j\}_{j<M}, \qquad \eta = e^{2\pi i/M},\ \zeta = e^{i\pi/M},$$

with radii constrained by $J_3 = \mu_3$. ($J_1 = J_2 = 0$ holds automatically.) So f_{eff} is, to all orders, the restriction of H to this curve.

Product formulas. Put $u_\pm = \rho_\pm^2$. Within one M-gon, $\sum_{i<j}\log|z_i - z_j|^2 = \binom{M}{2}\log u_\pm + \text{const}$. Between the two,

$$\sum_{j,k}\log|\rho_+\eta^j - \rho_-\zeta\eta^k|^2 = M\log\Big|\prod_{\omega^M=-1}(\rho_+ - \rho_-\omega)\Big|^2 = 2M\log\big(u_+^{M/2} + u_-^{M/2}\big),$$

using $\prod_{\omega^M=-1}(x - y\omega) = x^M + y^M$. With $\sigma_\pm = 1 + \lambda u_\pm$, this gives

$$H = -\tbinom{M}{2}(\log u_+ + \log u_-) - 2M\log\big(u_+^{M/2} + u_-^{M/2}\big) + (n-1)M(\log\sigma_+ + \log\sigma_-) + \text{const}.$$

Expansion on the constraint. The constraint is $g(u_+) + g(u_-) = 2g(1)$ with $g(u) = u/(1+\lambda u)$. Parametrise it by $g(u_\pm) = g(1) \pm \tau$. The swap $\tau \mapsto -\tau$ is a rotation by π/M, so H is even in τ: $H = H_0 + h_2\tau^2 + h_4\tau^4 + O(\tau^6)$. Direct expansion gives

$$h_2 = -\frac{M}{4}(1+\lambda)^2\big[M^2(1+\lambda)^2 - (4M-2)(1+\lambda^2)\big].$$

This vanishes exactly when $(1+\lambda^2)/(1+\lambda)^2 = M^2/(2(2M-1)) = n^2/(8(n-1))$, which is (2) for even n. The expansion also gives

$$h_4 = \frac{M(1+\lambda)^4}{96}\Big[M^4(1+\lambda)^4 - 16M^2(1+\lambda)(1+\lambda^3) + 24(2M-1)(1+\lambda^4)\Big].$$

The e-component of the displacement is $\varepsilon = \frac{\sqrt{n}}{2}(\rho_+ - \rho_-) + O(\rho-1)^2$, and $\tau = \frac{2}{(1+\lambda)^2}\,\frac{\rho_+ - \rho_-}{2} + O(\cdot)^2$. Since $h_2 = 0$, only the leading linear relation matters, and

$$c_4 = h_4\Big(\frac{2}{(1+\lambda)^2}\Big)^4 \frac{1}{n^2}.$$

Reducing modulo the quadratic relation (2) gives the stated rational function of n. The reduction was done symbolically in M and λ, by two independent scripts (Section 8). Positivity is clear: $n - 2 > 0$ and $m^4 + 40m^2 + 48m + 16 > 0$. □

4.3 Odd n: canonical coordinates, root-of-unity sums and residues

Theorem 7. *For odd $n \geq 5$ at $b = b_n$,*

$$c_4 = \frac{(n-1)(n-3)(n^2-9n+38)}{128\,n}.$$

In particular $c_4 > 0$, because $n^2 - 9n + 38$ has discriminant -71. Also $c_4(7) = 9/14$.

Proof. Canonical coordinates. Put $w = |z|^2/\sigma$ and $\vartheta = \arg z$. Then $\Omega = \sum_k dw_k \wedge d\vartheta_k$, because $2\,dx \wedge dy/\sigma^2 = du \wedge d\vartheta/\sigma^2$ with $u = |z|^2$, and $dw = du/\sigma^2$. In these coordinates

$$J_3 = \sum_k w_k, \qquad J_1 + iJ_2 = \sum_k \sqrt{P(w_k)}\, e^{i\vartheta_k}, \qquad P(w) = w(1-\lambda w),$$

$$\frac{|z_i - z_j|^2}{\sigma_i\sigma_j} = w_i + w_j - 2\lambda w_i w_j - 2\sqrt{P(w_i)P(w_j)}\cos(\vartheta_i - \vartheta_j).$$

So H has no one-body term, and J_3 is linear. The ring is $w_k = w_0 = 1/(1+\lambda)$, $\vartheta_k = \phi_k$.

Ansatz. Write $w_k = w_0 + \delta_k$ and $\vartheta_k = \phi_k + \psi_k$. On $J^{-1}(\mu)$ we have $\sum \delta_k = 0$ *exactly*, because J_3 is linear. Fix the gauge by $\sum \psi_k = 0$, and take the kernel direction $\delta_k = A\cos(l\phi_k)$, $l = (n-1)/2$. Since $2l \equiv -1 \pmod n$, a^2 lies in Fourier modes 0 and ± 1, so only these modes are slaved at order A^2. Mode 0 is removed, as just said. The reflection $k \mapsto -k$ fixes the kernel vector, so the slaved mode-1 part has the form

$$\delta_k = A\cos(l\phi_k) + A^2 p\cos\phi_k, \qquad \psi_k = A^2 s\sin\phi_k.$$

Corrections of order A^3 do not affect the A^4 coefficient. The partial gradient $\partial_b h_\mu$ at the $O(A^2)$-truncated slaved point is $O(A^3)$, and pairing it with an $O(A^3)$ correction gives $O(A^6)$. Modes $m \geq 2$ have no a^2 coupling, so their slaved amplitude is $O(A^3)$ and contributes $O(A^6)$ as well.

Constraint. $J_2 = 0$ holds by reflection symmetry. At order A^2, the condition $J_1 = 0$ reads

$$\tfrac{n}{2}A^2\big[Q'(w_0)p - Q(w_0)s + \tfrac{1}{4}Q''(w_0)\big] = 0, \qquad Q = \sqrt{P},$$

with $Q(w_0) = 1/(1+\lambda)$, $Q'(w_0) = (1-\lambda)/2$ and $Q''(w_0) = -(1+\lambda)^3/4$. Here we used $\sum_k \cos^2(l\phi_k)\cos\phi_k = n/4$. The inhomogeneous term $\frac{1}{4}Q''$ is the curvature of the level set, which in these coordinates sits in J_1 rather than J_3.

Sums over roots of unity. Expanding H along the ansatz requires sums over k of rational functions of $e^{i\phi}$ and $e^{il\phi}$ for offsets $\phi = 2\pi m/n$. Put $t = e^{i\pi m/n}$. Then $e^{i\phi} = t^2$ and $e^{il\phi} = (-1)^m/t$, and $\sum_{m=1}^{n-1}$ becomes half the sum over the $2n$-th roots of unity $t \neq \pm 1$. For each sign $\varsigma = t^n = \pm 1$, the sum $\sum_{t^n=\varsigma} R(t)$ is evaluated by residues of $R(t)\, nt^{n-1}/(t^n - \varsigma)$. The finite poles of R are only at $t = \pm 1$ (and possibly 0). Every piece has Laurent degree ≤ 0 at 0 and at ∞, so the residue at 0 vanishes and the residue at ∞ is $-n$ times the constant term of R at ∞. The residues at ± 1 are computed from binomial expansions of t^n about ± 1, which are polynomials in n. Every sum is therefore an explicit rational function of n and λ, valid for all odd $n \geq 5$. This is the step that replaces the numerical evaluation in [7].

Result. The A^2 coefficient of H along the kernel is

$$f_2 = -\frac{n(n-1)(1+\lambda)^2}{64}\big[(n-7)(1+\lambda^2) + 2(n+1)\lambda\big],$$

which vanishes exactly at b_n; for odd n, (2) reads $(1+\lambda^2)/(1+\lambda)^2 = (n+1)/8$. The A^3 coefficient vanishes. The A^4 coefficient is $E(p,s) = E_{00} + E_{10}p + E_{01}s + E_{20}p^2 + E_{02}s^2$ (there is no ps term), with

$$\begin{aligned} E_{20} &= \tfrac{1}{16}n(n-1)(1-\lambda)^2(1+\lambda)^2, & E_{02} &= \tfrac{1}{4}n(n-1), \\ E_{10} &= \tfrac{1}{64}n(n-1)(1-\lambda)(1+\lambda)^3\big[(n-5)(1+\lambda^2) + 2(n-1)\lambda\big], & E_{01} &= \tfrac{1}{64}n(n-1)^2(1+\lambda)^4, \end{aligned}$$

$$\begin{aligned} E_{00} = \tfrac{n(n-1)(1+\lambda)^4}{8192}\Big[&(n^3+n^2-69n+291)(1+\lambda^4) \\ &+ 4(n^3+n^2-21n-13)(\lambda+\lambda^3) + (6n^3+6n^2-30n+82)\lambda^2\Big]. \end{aligned}$$

Eliminate s with the J_1 constraint. The coefficient of p^2 in what remains is $a_2 = \frac{1}{8}n(n-1)(1-\lambda)^2(1+\lambda)^2 > 0$, so the slaved p is a minimum. Normalising $\delta_k = A\cos(l\phi_k)$ to the unit vector e gives $A = \varepsilon\,(2/n)^{1/2}\,2/(1+\lambda)^2$, so

$$c_4 = \frac{64}{n^2(1+\lambda)^8}\min_p E\big(p, s(p)\big).$$

Reducing modulo the threshold relation $(n-7)(1+\lambda^2) + 2(n+1)\lambda = 0$ gives the stated formula for $n \neq 7$. The denominator does not vanish modulo the relation unless $n = 7$. Direct substitution $\lambda = 0$, $n = 7$ gives $9/14$, which agrees with the formula. □

Remark 8 (Where the difficulty lies). Once the closed forms are known, $c_4 > 0$ is immediate. For even n every term of $(n-2)(m^4 + 40m^2 + 48m + 16)$ is positive for $m = n-2 \geq 2$. For odd n, $n^2 - 9n + 38$ has discriminant -71, and $(n-1)(n-3) > 0$ for $n \geq 5$. The content of Theorems 6 and 7 is the derivation of c_4, in particular the evaluation of the root-of-unity sums for general n, which [7, p. 23] could not carry out in closed form. The odd formula vanishes at $n = 3$, because of the factor $n - 3$. This is consistent with $n = 3$ being a different situation (bifurcation at the equator, $b = 1$), which needs, and in [7] receives, separate treatment; see the remark after the Corollary in Section 2.

Remark 9 (What c_4 is, for odd n). For odd n, c_4 is not the quartic coefficient of H along the kernel. It is the quartic coefficient of the reduced Hamiltonian after the mode-1 direction has been eliminated: the amplitude p (radial) and s (tangential) of the slaved mode-1 displacement are fixed by minimisation and by the constraint $J_1 = 0$. Because c_4 is computed on the level set $J^{-1}(\mu)$, it also contains the curvature term $\frac{1}{4}Q''(w_0)$ in the J_1 constraint.

Remark 10. The odd-n derivation was done symbolically in n and λ by a second AI reader. It was then audited step by step by a third reader: the coordinate change, the decoupling, completeness of the ansatz (including the absence of aliasing at $n = 5, 7$), the slaved modes, the J_1 constraint, the residues and Laurent degrees, and the invariant theory. The third reader also checked it by an independent exact route, described next.

4.4 Exact values and independent verification

Table 1 lists c_4 for $n = 4, \ldots, 25$ from the closed forms. Every value has been checked independently of the derivations above. The check is an order-by-order KKT expansion of the level-set minimisation in *Cartesian* coordinates, with all $2n$ coordinates unknown and no Fourier ansatz, no symmetry reduction and no residue sums. It is carried out exactly over $\mathrm{GF}(p)$ for three primes $p \approx 2^{61}$ with $p \equiv 1 \pmod{4n}$, at both algebraic conjugates of b_n. In every case the computation confirms three things exactly: the kernel eigenvalue vanishes, the cubic term vanishes, and c_4 equals the tabulated rational number. A deliberately perturbed value is rejected (negative control). Agreement modulo three primes of size 2^{61} is overwhelming evidence, though strictly it is not a proof over $\mathbb{Q}$. The reduction from the coefficients stated in the proofs to every table entry is, however, verified exactly over $\mathbb{Q}$ in Lean (see Section 8). The proofs are Theorems 6 and 7. The same engine in 40-digit floating point, with the real root b_n, agrees to 10^{-37} or better. A separate 40-digit constrained minimisation in the original chart gives the same values to 15 digits or better for $n = 4, \ldots, 13, 15, 17, 19, 21, 25$.

5 Nonlinear stability for a degenerate minimum

The classical energy–momentum criteria assume a definite Hessian, a compact group, or both. Here the Hessian is only semidefinite, and the group is $SE(2)$ or $PSL(2, \mathbb{R})$ for $n \geq 7$. We therefore give a self-contained argument. It uses no stability theorem from the literature. The construction (a cross-section, then an energy–momentum function with a penalty $(J_3 - \mu_3)^2$) is classical in spirit, and we claim no novelty for it.

Abstract setting. Let (P, Ω) be a symplectic manifold and H a smooth function. A Lie group G acts by symplectomorphisms preserving H. Functions J_1, J_2, J_3 are Hamiltonians for generators ξ_1, ξ_2, ξ_3, where ξ_3 generates a circle action $K \cong SO(2)$, and $\mathrm{span}(\xi_1, \xi_2)$ is Ad_K-invariant. Let x_0 be a relative equilibrium, $dF(x_0) = 0$ for $F = H - \omega_0 J_3$. Fix a K-invariant Riemannian metric on P. Assume:

(A1) $J_1(x_0) = J_2(x_0) = 0$, K rotates the pair (J_1, J_2), and $\{J_1, J_2\}(x_0) \neq 0$.

(A2) K acts freely near x_0.

(A3) There is a smooth curve $j \mapsto x_j$ of relative equilibria through x_0 with $J_1 = J_2 = 0$ and $J_3(x_j) = \mu_3 + j$, with smooth angular velocities $\omega(j)$ and $\omega(0) = \omega_0$.

n	b_n	geometry	c_4 (level set)	decimal
4	0.2679491924	sphere	1/4	0.25
5	0.1715728753	sphere	9/40	0.225
6	0.0557280900	sphere	23/30	0.7666667
7	0	plane	9/14	0.6428571
8	−0.0627460668	H^2	95/56	1.6964286
9	−0.1010205144	H^2	19/12	1.5833333
10	−0.1428571429	H^2	49/15	3.2666667
11	−0.1715728753	H^2	75/22	3.4090909
12	−0.2024108640	H^2	755/132	5.7196970
13	−0.2251482266	H^2	675/104	6.4903846
14	−0.2492775954	H^2	1693/182	9.3021978
15	−0.2679491924	H^2	56/5	11.2
16	−0.2875950927	H^2	1141/80	14.2625
17	−0.3033370453	H^2	609/34	17.9117647
18	−0.3197938296	H^2	3190/153	20.8496732
19	−0.3333333333	H^2	27	27
20	−0.3474164357	H^2	11139/380	29.3131579
21	−0.3592455180	H^2	2175/56	38.8392857
22	−0.3714998241	H^2	6145/154	39.9025974
23	−0.3819660113	H^2	2475/46	53.8043478
24	−0.3927725152	H^2	29183/552	52.8677536
25	−0.4021298312	H^2	7227/100	72.27

Table 1: Level-set effective quartic c_4 (Definition 5; $r_0 = 1$, H as in (1), Euclidean-unit kernel vector). All 22 rational values were confirmed by the exact GF(p) Cartesian KKT computation.

(A4) On $M_\mu = J^{-1}(\mu)/K$, the reduced Hamiltonian h has positive semidefinite Hessian at $[x_0]$, with kernel K_0 and positive definite on a complement E. The function $g(a) = h(a, \beta(a)) - h(0)$ (β the slaving map) has vanishing cubic Taylor term and positive definite quartic Taylor term q_4.

Each J_i is conserved, since $\{H, J_i\} = dH(\xi_i) = 0$.

Lemma 11 (Lemma A: cross-section)**.** *Near x_0, $S := \{J_1 = J_2 = 0\}$ is a K-invariant symplectic submanifold of codimension 2. It is invariant under the flow of H, and the flow on S is the Hamiltonian flow of $H|_S$. There is a K-invariant neighbourhood U of the whole orbit $K \cdot x_0$, and a continuous map $\gamma : U \to G$ with $\gamma(x) \cdot x \in S$, $\gamma|_{K \cdot x_0} = e$ and $\gamma(kx) = k\gamma(x)k^{-1}$ for $k \in K$.*

Proof. Since $\{J_1, J_2\}(x_0) \neq 0$, dJ_1 and dJ_2 are independent. Moreover $TS^\Omega = \operatorname{span}(X_{J_1}, X_{J_2})$ with $\Omega(X_{J_1}, X_{J_2}) \neq 0$, so TS^Ω and TS are complementary and $\Omega|_S$ is nondegenerate. Because J_1, J_2 are conserved, X_H is tangent to S, and $\Omega|_S(X_H, v) = dH(v)$ for $v \in TS$ gives $X_H|_S = X_{H|_S}$. S is K-invariant because K rotates (J_1, J_2).

For γ, let $\Psi(x, s_1, s_2) = (J_1, J_2)(\exp(s_1\xi_1 + s_2\xi_2) \cdot x)$. At $(x_0, 0)$, its derivative in (s_1, s_2) is the matrix $dJ_i(\xi_j(x_0)) = \pm\{J_i, J_j\}(x_0)$, which is invertible by (A1). The implicit function theorem gives $\varrho > 0$, a neighbourhood U_0 of x_0, and a smooth $s(x)$ on U_0, such that $s(x)$ is the *unique* solution of $\Psi(x, s) = 0$ with $|s| < \varrho$, and $s(x_0) = 0$. Put $\gamma(x) = \exp(s_1(x)\xi_1 + s_2(x)\xi_2)$.

To extend to the whole orbit, note the following. For $k \in K$, $k \exp(s \cdot \xi)k^{-1} = \exp(\operatorname{Ad}_k(s \cdot \xi))$ with $\operatorname{Ad}_k(s \cdot \xi) \in \operatorname{span}(\xi_1, \xi_2)$, and $|\operatorname{Ad}_k s| = |s|$ for a K-invariant norm. Also S is K-invariant. So for $x \in kU_0$, the element $k\gamma(k^{-1}x)k^{-1}$ is the unique element $\exp(s' \cdot \xi)$ with $|s'| < \varrho$ carrying x into S. This uniqueness shows that the definitions on overlapping translates $kU_0 \cap k'U_0$ agree.

Hence γ is well defined and continuous on the K-invariant neighbourhood $U = K \cdot U_0$, and it is K-equivariant. □

Lemma 12 (Lemma B1: quartic lower bound on M_μ)**.** *Under (A4), in coordinates $w = (a, b) \in K_0 \times E$ centred at $[x_0]$, there are $\kappa, \varrho > 0$ with*

$$h(a,b) - h(0) \;\geq\; \kappa\big(|a|^4 + |b|^2\big) \qquad \text{for } |w| < \varrho.$$

Proof. Let $f = h - h(0)$, so $d^2 f(0) = \operatorname{diag}(0, B)$ with $B > 0$. By the implicit function theorem there is a smooth β with $\partial_b f(a, \beta(a)) = 0$ and $\beta(0) = 0$. Moreover $d\beta(0) = 0$, because the cross block vanishes, so $|\beta(a)| \leq L|a|^2$. Taylor's theorem in b about $\beta(a)$ gives $f(a,b) - g(a) \geq \frac{1}{4}\lambda_{\min}(B)|b - \beta(a)|^2$ for small (a, b).

Next, $f(a, \beta(a)) - f(a, 0) = O(|a|^2) \cdot O(|a|^2)$, since $\partial_b f(a, 0) = O(|a|^2)$ and $\beta = O(|a|^2)$. So the Taylor terms of g of order ≤ 3 are those of $f|_{K_0}$. These vanish: order 2 by (A4) and order 3 by assumption. Hence $g(a) \geq \frac{1}{2}\min_{|a|=1} q_4\,|a|^4$ for small a. Finally $|b|^2 \leq 2|b - \beta|^2 + 2L^2|a|^4$. □

Lemma 13 (Lemma B2: uniformity in nearby momentum)**.** *By (A2), S/K is a manifold near $[x_0]$. Choose coordinates (j, w) on it with $j = J_3 - \mu_3$; this is possible because $dJ_3 \neq 0$ on S/K, since $X_{J_3} \neq 0$ and $\Omega|_S$ is nondegenerate. Choose them so that $\{w = 0\}$ is the curve $[x_j]$ of (A3), and $\{j = 0\} = M_\mu$ with the coordinates of Lemma 12. Let $\Phi(j, w) = H(j, w) - H(j, 0)$. Then there are $C, \varrho > 0$ with*

$$\Phi(j,w) \;\geq\; \tfrac{\kappa}{2}\big(|a|^4 + |b|^2\big) - Cj^2 \qquad (|j|, |w| < \varrho).$$

Proof. We have $\Phi(j, 0) = 0$. Also $\partial_w \Phi(j, 0) = 0$ for every j, because x_j is a critical point of H on $\{J_3 = \mu_3 + j\} \cap S$. Differentiating these identities in j gives $\partial_j \Phi(0, 0) = 0$ and $\partial_w \partial_j \Phi(0, 0) = 0$. Hence $|\partial_j \Phi(0, w)| \leq L'|w|^2 = L'(|a|^2 + |b|^2)$. Note that there are no terms linear in a or b multiplied by j. Taylor's theorem in j gives

$$\Phi(j,w) \geq \Phi(0,w) - L'|j|\,(|a|^2 + |b|^2) - C_2 j^2.$$

For $\Phi(0, w) = h(w) - h(0)$ use Lemma 12. The term $j|a|^2$ is handled by AM–GM: with $X = \sqrt{\kappa}\,|a|^2$ and $Y = L'|j|/\sqrt{\kappa}$,

$$L'|j|\,|a|^2 = XY \leq \tfrac{1}{2}X^2 + \tfrac{1}{2}Y^2 = \tfrac{\kappa}{2}|a|^4 + \tfrac{L'^2}{2\kappa}j^2.$$

The term $j|b|^2$ is absorbed directly: $L'|j|\,|b|^2 \leq \frac{\kappa}{2}|b|^2$ once $|j| \leq \kappa/(2L')$. Altogether $\Phi \geq \frac{\kappa}{2}(|a|^4 + |b|^2) - (L'^2/(2\kappa) + C_2)j^2$. □

Remark 14 (The minimum does not persist at nearby momentum). Lemma 13 only bounds how deep Φ can go below zero; it does not say $\Phi \geq 0$. In fact the strict minimum does *not* persist. On the unstable side ($j < 0$, slightly smaller ring), the kernel eigenvalue of the reduced Hessian at $\mu_3 + j$ is negative and of order $|j|$. The ring is then a saddle of the reduced Hamiltonian, and the bifurcated minima sit at amplitude of order $|j|^{1/2}$ with depth of order j^2.

We computed this numerically on the $D_{n/2}$-invariant family. For $n = 8$, $\min \Phi/j^2$ equals -0.0022841, -0.0022845 and -0.0022845 at $j = -10^{-2}, -10^{-3}, -10^{-4}$, and the minimiser satisfies $\tau^*/\sqrt{|j|} \to 0.1542$. For $n = 12$, $\min \Phi/j^2 \to -0.013679$ and $\tau^*/\sqrt{|j|} \to 0.2007$. On the stable side ($j > 0$) the minimum is exactly 0, attained at the ring.

So there is no neighbourhood on which the reduced Hamiltonians at all nearby momenta have uniform strict minima. The step in Morse-lemma-based proofs that relies on such uniformity is therefore not available here. The penalty function in Lemma 15 replaces it.

Lemma 15 (Lemma B3: conserved function with a strict minimum modulo K)**.** *On S let $V = H - \omega_0 J_3 + K_c(J_3 - \mu_3)^2$. Then V is conserved and K-invariant. For K_c large enough, the function it induces on S/K has a strict local minimum at $[x_0]$.*

Proof. Write $V(j,w) - V(0,0) = \Phi(j,w) + \psi(j) + K_c j^2$ with $\psi(j) = H(x_j) - H(x_0) - \omega_0 j$. Along the family, $d(H - \omega(j)J_3)(x_j) = 0$ gives $\frac{d}{dj}H(x_j) = \omega(j)$. So $\psi(0) = \psi'(0) = 0$ and $|\psi(j)| \le C_3 j^2$. By Lemma 13,

$$V - V(0,0) \ge \tfrac{\kappa}{2}(|a|^4 + |b|^2) + (K_c - C - C_3)j^2,$$

which is positive for $(j,w) \neq 0$ once $K_c > C + C_3$. □

Lemma 16 (Lemma B4: Lyapunov stability on S)**.** *The orbit $K \cdot x_0$ is Lyapunov stable for the flow on S.*

Proof. $K \cdot x_0$ is compact, and $V - V_0 > 0$ on a punctured K-invariant tubular neighbourhood N_r of $K \cdot x_0$ in S (Lemma 15). Given $\varepsilon < r$, let $m = \min\{V - V_0 : \ x \in S, \ \mathrm{dist}(x, K \cdot x_0) = \varepsilon\}$. This is a minimum over a compact set, so $m > 0$. Choose δ such that $V - V_0 < m$ on the δ-neighbourhood. Since V is conserved, a trajectory starting there never reaches distance ε. It stays in a compact, collision-free set, so it exists for all time. □

Theorem 17. *Under (A1)–(A4), x_0 is Lyapunov stable modulo K in the full phase space P, with no restriction on the momentum of the perturbation.*

Proof. Let $x(0)$ be close to $K \cdot x_0$, put $\gamma = \gamma(x(0))$ (Lemma 11), and let $y(t) = \gamma \cdot x(t)$. Since G acts symplectically and preserves H, it commutes with the flow, so $y(t)$ is the trajectory through $y(0) = \gamma \cdot x(0) \in S$. Moreover $y(0) \to K \cdot x_0$ and $\gamma \to e$ as $x(0) \to K \cdot x_0$. By Lemma 16, $y(t)$ stays within ε' of $K \cdot x_0$. Then

$$\mathrm{dist}(x(t), K \cdot x_0) \le d(\gamma^{-1} y(t), y(t)) + \mathrm{dist}(y(t), K \cdot x_0) \le \omega(\gamma) + \varepsilon',$$

where $\omega(\gamma) = \sup\{d(\gamma^{-1}y, y)\}$ over a fixed compact neighbourhood of $K \cdot x_0$. This tends to 0 as $\gamma \to e$ by continuity of the action. □

Proof of Theorem 1. We verify (A1)–(A4) for the ring at b_n, $n \ge 4$, in all three geometries.

- (A1): $J_+(x_0) = 0$ by $\mathbb{Z}_n$ symmetry, and a rotation by ψ multiplies J_+ by $e^{i\psi}$. $\{J_1, J_2\}(x_0) = n(1 - b_n)/(2(1 + b_n)) \neq 0$ by (4), since $|b_n| < 1$. The span of ξ_1, ξ_2 is Ad_K-invariant in $\mathfrak{so}(3)$, $\mathfrak{se}(2)$ and $\mathfrak{sl}(2, \mathbb{R})$.
- (A2): a nontrivial rotation moves a labelled ring.
- (A3): the concentric rings of radius r are relative equilibria for every r [7, (2.1)], with $J_+ = 0$, and $J_3 = nr^2/(1 + \lambda r^2)$ is strictly increasing in r.
- (A4): Proposition 4 gives the Hessian and its kernel. By the symmetry argument after Definition 5, $q_4 = c_4|a|^4$ in a chart, and $c_4 > 0$ by Theorems 6 and 7.

Theorem 17 now gives Theorem 1. □

6 Relation to the literature

Patrick–Roberts–Wulff. The following is an independent confirmation for the sphere and the hyperbolic plane. Patrick, Roberts and Wulff [8] consider a free action with Ad*-equivariant momentum map. Their Theorem 4.2 states that p_e is G-stable if $H^{-1}(H(p_e)) \cap J^{-1}(T_2^{U_{\mu_e}}(\mu_e)) \cap U_S = \{p_e\}$ for some neighbourhoods U_S (in a slice) and U_{μ_e}. No Hessian condition is required. When $\mathfrak{g}^*/G$ is Hausdorff at μ_e, which holds if μ_e is regular [8, Prop. 4.10(1)], this condition is equivalent to p_e being isolated in its energy level within the reduced space. A strict local minimum (Lemma 12) is isolated. Their Corollary 5.4 then upgrades G-stability to $G_{\mu_e}^o$-stability whenever there is a $G_{\mu_e}^o$-invariant inner product on $\mathfrak{g}^*$, which is automatic here since $G_\mu = SO(2)$ is compact.

For $SO(3)$ ($\mu \neq 0$) and $PSL(2,\mathbb{R})$ (μ timelike), μ is regular, and the action is free and proper on a neighbourhood of the labelled ring. With J shifted to its equivariant form (for $\lambda \neq 0$), these results therefore give G_μ-stability of the ring at b_n for $n \neq 7$, once the strict minimum of the reduced Hamiltonian is known, i.e. once Sections 3 and 4 are in hand.

The plane. For $\lambda = 0$ the momentum map of $SE(2)$ is equivariant only for an affine action (equation (4): $\{J_1, J_2\} = n/2$ is a cocycle). In the central extension that makes it equivariant, the centre acts trivially, so the action is not free. So [8] does not apply verbatim, and the self-contained Theorem 17 is what covers $n = 7$.

Montaldi 1997. Montaldi–Tokieda's Proposition 9 builds on [5] (cited as "cf. [17]" in its proof [7, p. 20], and as "[17, Theorem 1.2]" on p. 10), whose setting is a compact group, as for the sphere. We have read only that paper's abstract and the way it is quoted in [7, pp. 10, 20], and we do not rely on it. Montaldi and Rodríguez-Olmos [6] treat noncompact groups under formal stability (a definite Hessian on the symplectic normal space). Since the present case combines the groups $SE(2)$ and $PSL(2,\mathbb{R})$ with a degenerate (quartic) minimum, Section 5 gives a self-contained argument that covers all of them at once.

7 Numerical confirmation

We integrated the $n = 8$ ring in the hyperbolic plane directly. This is a check on the theory, not part of the proof. The code is independent of that used for c_4. It uses the Poincaré disk with area form $4\,dx \wedge dy/(1-|z|^2)^2$, DOP853 with relative tolerance 10^{-12}, and final time $T = 2000$, which is more than six periods of the nonlinear oscillation at amplitude 10^{-4}.

The initial data are the ring at $\xi = \xi^*(1 + \mathrm{rel})$, where $\xi = |z|^2$ and $\xi^* = -b_8$ is the threshold, plus the critical mode $|z_k| = \sqrt{\xi}(1 + \delta(-1)^k)$. Here rel > 0 is the stable side. The diagnostic is $D(t) = \max_{i<j} |d_{ij}(t) - d_{ij}^{\mathrm{ring}}|$ in hyperbolic distance, which is invariant under the whole group. Energy and momentum are conserved to 10^{-8} and $6 \cdot 10^{-10}$, and the unperturbed ring drifts by less than $6 \cdot 10^{-12}$.

$\max_t D(t)/D(0)$	rel $= -10^{-2}$	-10^{-3}	0	$+10^{-3}$	$+10^{-2}$
$\delta = 10^{-3}$	29.34	9.13	1.000	1.000	1.000
$\delta = 10^{-4}$	293.6	91.86	1.000	1.000	1.000

Below threshold the deviation saturates at an amplitude independent of δ: about 0.029 for rel $= -10^{-2}$ and 0.0091 for rel $= -10^{-3}$. Their ratio is close to $\sqrt{10}$. This is the $\sqrt{\Delta}$ signature of a supercritical pitchfork: the orbit wanders near the stable bifurcated pair of $n/2$-gons. At threshold

and above it, $D(t)$ never exceeds $D(0)$. It oscillates, spends 38–42% of the time below $D(0)/2$ at threshold, and shows no secular growth. If the quartic were negative, threshold runs would leave the δ-neighbourhood on the nonlinear time scale. The same behaviour was observed for the image ("tanh") hyperbolic model, which is not part of Conjecture 13.

8 Limitations, status and computer-assisted parts

Review status. This draft has been checked only by independent AI readers (separate model instances that re-derived the results with their own code before reading the candidate argument) and by computer calculations. **No human specialist has reviewed it.** Its claims should be treated accordingly until such a review has taken place.

Literature not read. We have not read Kurakin–Yudovich [4] beyond its abstract, nor Kurakin [2, 3]. We make no claim that our results agree with, extend, or differ from theirs. One unverified observation: if $H = -\sum \log|z_i - z_j|$ and a per-vortex radial amplitude are used, our level-set value at $n = 7$ rescales to $\frac{9}{14} \cdot \frac{49}{8} = \frac{63}{16}$. The number 63/16 has been attributed to [4] as the heptagon's quartic coefficient in secondary notes we started from, but we have not verified that attribution or the normalisation used there. This is therefore an unverified consistency remark, not a claim of agreement. Kurakin's sphere results may already cover $n = 4, 5, 6$ at the critical latitude.

What is computer-assisted. All computations are exact symbolic (sympy) or exact modular unless labelled numerical. Paths are relative to `conj13/` in the accompanying repository.

- *Even-n closed form* (Theorem 6). The expansion of H on $\mathrm{Fix}(D_M) \cap J^{-1}(\mu)$, symbolic in M and λ, and its reduction modulo (2) were done twice, independently: `even_exact.py`, `even_formula.py` and `review/even_check.py`. The product formulas were checked by resultants for $M \le 6$.
- *Odd-n closed form* (Theorem 7). The symbolic derivation in n and λ, including the residue evaluation of root-of-unity sums with Laurent-degree checks, is in `review/odd_general.py`, `review/odd_general_run.py` and `review/odd_assemble.py`. The human-readable steps are reproduced above. The output coefficients f_2, E_{ij} and a_2 are stated in the proof so that they can be re-derived by hand or with another system.
- *Independent exact check of Table 1.* Cartesian order-by-order KKT over $\mathrm{GF}(p)$, with no ansatz: `review3/kkt_exact.py` and `review3/run_exact.py` (odd n and $n = 8$), and `paper/table_check.py` (all $n = 4, \ldots, 25$, with a negative control; output in `paper/table_check.out`).
- *Spectral checks* (Proposition 4, numerical, 40 digits): `gaps/spectral_check.py`, plus the bordered inertia in `review3/hess_check.py`.
- *Lemma 15 sanity check* (numerical): `gaps/review/indep_check.py`. For $n = 4, \ldots, 9$, the minimum of $V - V_0$ on spheres of radius ε in S scales as $c_4\varepsilon^4$.
- *Remark 14*: `gaps/nearby_momentum.py`.
- *40-digit constrained minimisation* (numerical, Table 1): `engine.py`, `reduced_quartic*.py`, `review/c4_levelset.py`.

- *Integrations*: `integrate.py` with outputs in `out/`.

The parts of the proof that use a computer are the algebraic simplifications in Theorems 6 and 7: series expansion, residue sums, and reduction modulo a quadratic relation. Each was reproduced by an independent route. The structural steps (Proposition 4, Section 5) are pen-and-paper.

Formal verification (Lean 4). The exact-algebra layer is formalised in Lean 4 with Mathlib, in `lean/Havelock/Conj13Algebra.lean`, with no unproved steps and only the standard axioms. The repository's continuous-integration build checks the Lean files. The precise scope is as follows. The coefficients stated in the proofs of Theorems 6 and 7, namely h_2, h_4 (even n) and f_2, $E_{00}, E_{10}, E_{01}, E_{20}, E_{02}$, a_2 (odd n), are explicit closed-form rational expressions in n (or $M = n/2$) and $b = \lambda r_0^2$. Lean takes these expressions as definitions and proves, symbolically in n and b, everything downstream of them:

- that h_2 and f_2 vanish exactly on the threshold relation (2);
- that eliminating s by the J_1 constraint and minimising over p (odd n), and reducing modulo the threshold relation (both parities), gives the stated closed forms for c_4, for every n;
- that these equal c_4 at the true (irrational) b_n and are positive for every $n \geq 4$, and every entry of Table 1, exactly over $\mathbb{Q}$.

It also proves the threshold facts (b_n in closed form, its uniqueness in $(-1, 1]$ and its sign pattern, for all $n \geq 4$), the spectral algebra of Proposition 4 (including the Fejér identity for all n and the eigenvalues $3/2$ and $8/3$ at $n = 4, 5$). A second layer, `lean/Havelock/Conj13Series.lean`, derives from eq. (1) the even-n coefficients h_2, h_4 along the symmetric curve on the level set (and the limit $(H - H_0)/\varepsilon^4 \to c_4$ at b_n), the odd-n coefficients f_2 and E_{ij} (the full fourth-order expansion along the ansatz, with no A^3 term, via the first- and second-order Fejér identities), the J_1 constraint along the ansatz, and both normalisation factors relating τ and A to ε. What Lean does *not* check are the structural steps: symmetric criticality (even n), completeness of the odd ansatz, and that the slaved minimum is the quartic coefficient of the reduced Hamiltonian. These rest on the arguments of §§3–4 and on the exact GF(p) Cartesian computation described above. For Section 5, `lean/Havelock/Conj13Stability.lean` proves Lemmas B1–B4 and Theorem 17's argument in an abstract setting (a group acting on a metric space, the chart and the slice taken as hypotheses); its instantiation to the vortex ring is not formalised.

Scope. The result concerns the family (1) of [7], i.e. the "sinh" model in the hyperbolic case. It says nothing about the image ("tanh") hyperbolic model or the flat disk with a wall. The stability obtained is Lyapunov stability modulo $SO(2)$. We do not address stability on exponentially long times for perturbations of the full system, KAM-type questions, or the bifurcated branches beyond what [7, Remark 11] states.

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